

Software Requirements Specification (SRS)

Project: Automated Student Financial Clearance System
University of Eswatini (UNESWA)
Standard: IEEE Std 830-1998

1. Introduction

1.1 Purpose

The purpose of this document is to provide a complete description of all the software requirements for the Automated Student Financial Clearance System for the University of Eswatini.

1.2 Scope

This project aims to digitize and automate the financial clearance process to reduce costs and time consumption for both students and the institution.

1.3 Definitions, Acronyms, and Abbreviations

- SRS: Software Requirements Specification document which outlines the requirements of the project.
- UNESWA: University of Eswatini, the primary institution for which the system is being developed.
- Moodle: The existing e-learning platform that will be synchronized with the new clearance system.

1.4 References

- IEEE Std 830-1998 standard for SRS development.
- Eswatini Data Protection Act regarding student privacy.
- UNESWA ICT Governance Framework for project compliance.

2. Overall Description

2.1 Product Perspective

The system serves as a standalone automation tool that is deeply integrated with the university's Moodle student services section.

2.2 Product Functions

- The system shall allow students to log in using their university credentials.
- The application shall provide an interface for students to upload digital copies of their financial documents.
- The system shall automatically synchronize student clearance status with the Finance Office database.

2.3 User Characteristics

- Students: Users will interact with the system to submit documents and view their clearance progress.
- Finance Officers: Administrators will use the system to monitor submissions and verify complex cases.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces: The user interface shall be web-based and responsive, featuring masked fields for password security.

3.1.2 Hardware Interfaces: The system must be hosted on servers capable of handling 5,000 concurrent student connections.

3.1.3 Software Interfaces: The software will interface with MySQL or MongoDB databases for record management.

3.2 Functional Requirements

3.2.1 Data Entry and Validation

- The system shall require the user to provide a valid Student Identification Number and a unique password.
- The system shall require the student to upload a valid Proof of Payment in an approved digital format.
- The application shall permit the student to upload their current Financial Statement for office review.

3.2.2 Security and Risk Mitigation

- The system shall enforce strong password requirements to prevent personality theft and unauthorized access.
- The software shall use high-resolution verification standards to detect and prevent

document fraud.

- The system shall implement size limits and compression options to prevent system overload from large file uploads.

3.3 Performance Requirements

- The system shall support up to 5,000 concurrent logins to ensure stability during peak registration periods.
- The system shall deliver a seamless user experience by maintaining high response speeds across all campus platforms.

3.4 Design Constraints

- All application code must be written in JavaScript, C++, or Python to align with university standards.
- All source code must be stored in a private repository following a strict Git-flow branching model.

4. Project Timeline and Milestones

- Task A: Contract signing will be completed within 3 days between 4 May and 6 May.
- Task B: User analysis via questionnaires will be conducted between 7 May and 13 May.
- Task C: The system design phase will occur over 10 days, concluding on 23 May.
- Task D: Implementation and coding will take 15 days, ending on 7 June.
- Task E: System testing and security audits will be conducted between 8 June and 14 June.
- Task F: Evaluation and full deployment are scheduled for 15 June to 28 June.